



Barker College

2011 TRIAL HIGHER SCHOOL CERTIFICATE

Mathematics Extension 2

Staff Involved:

- GDH
- MRB
- BHC*
- VAB*

**PM THURSDAY 4TH AUGUST
TIME: 3 HOURS**

50 copies

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Make sure your Barker Student Number is on ALL pages of your answer sheets.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.

Total marks – 120

- Attempt Questions 1–8.
- ALL necessary working should be shown in every question.
- Start each question on a NEW page.
- Write on one side only of each answer page.
- Marks may be deducted for careless or badly arranged work.

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Total marks – 120

Attempt Questions 1– 8

Answer each question on a **SEPARATE** sheet of paper

			Marks
Question 1	(15 marks)	[START A NEW PAGE]	
(a)	(i)	Find $\int \frac{dx}{3 - 2x - x^2}$ using partial fractions.	4
	(ii)	Hence, or otherwise find $\int \frac{2 + x}{3 - 2x - x^2} dx$	2
(b)	(i)	Find $\int \frac{dx}{\sqrt{3 - 2x - x^2}}$	2
	(ii)	Hence, or otherwise, find $\int \frac{1 + 2x}{\sqrt{3 - 2x - x^2}} dx$	3
(c)		Find $\int \sqrt{x^2 + a^2} dx$ using integration by parts.	4

End of Question 1

Question 2 (15 marks) **[START A NEW PAGE]**

(a) (i) Solve $z^3 = \sqrt{2} + \sqrt{2}i$, giving answers in the form $R \operatorname{cis} \theta$. 2

(ii) Hence prove that $\cos\left(\frac{\pi}{12}\right) + \cos\left(\frac{9\pi}{12}\right) + \cos\left(\frac{-7\pi}{12}\right) = 0$ 1

(b) Find the locus of Z for the following:

You may give your answer as an equation or a graph, whichever you prefer.

(i) $\frac{Z-i}{Z-2}$ is purely real. 2

(ii) $\frac{Z-i}{Z-2}$ is purely imaginary. 2

(c) Let $z = \cos \theta + i \sin \theta$.

(i) Using de Moivre's Theorem and the Binomial Theorem, show that

$$\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$$
 3

(ii) Hence solve:

$$32x^5 - 40x^3 + 10x = 1$$
 3

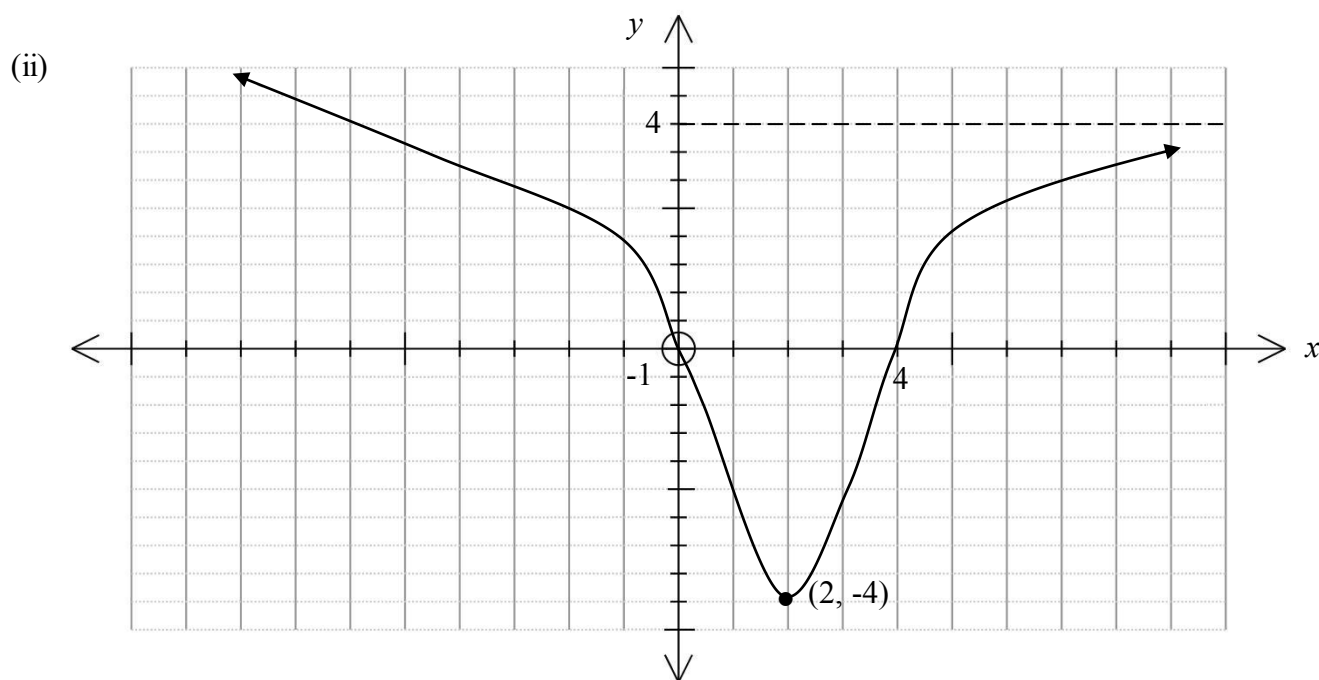
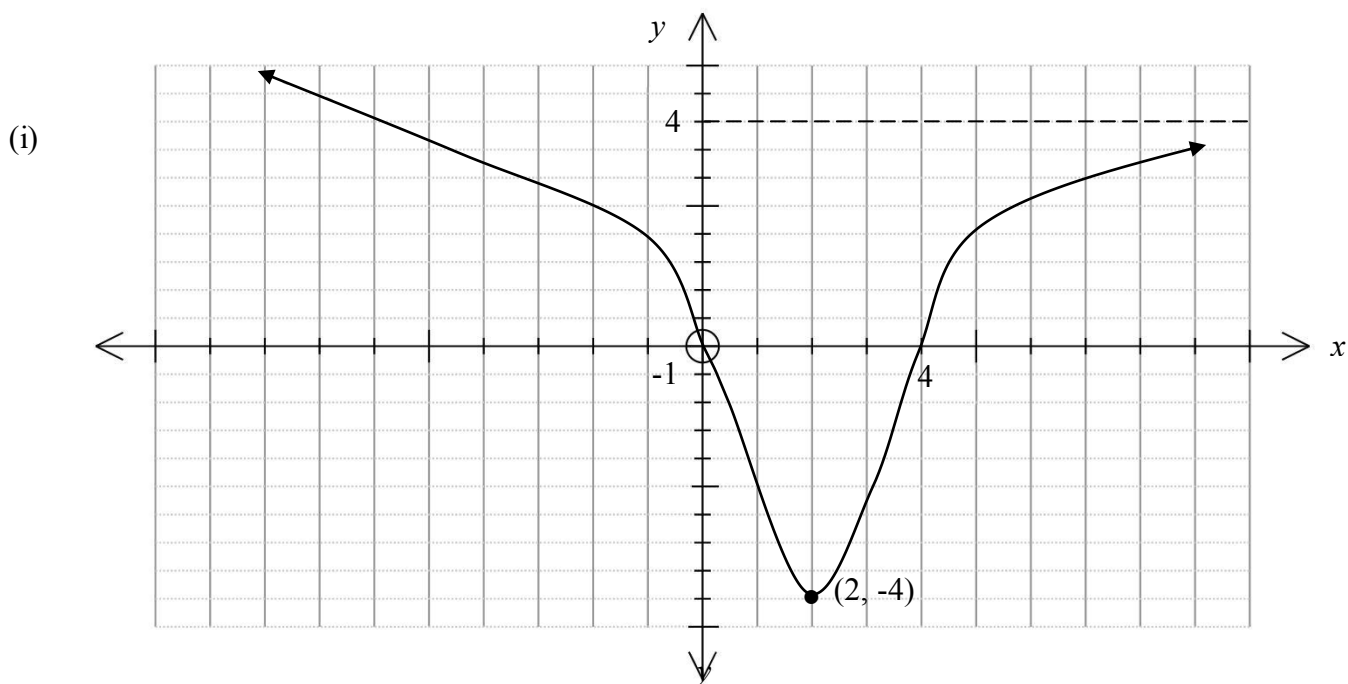
(iii) Hence prove that:

$$\cos\left(\frac{\pi}{15}\right) \cdot \cos\left(\frac{7\pi}{15}\right) \cdot \cos\left(\frac{11\pi}{15}\right) \cdot \cos\left(\frac{13\pi}{15}\right) = \frac{1}{16}$$
 2

End of Question 2

Question 3 (15 marks) **[START A NEW PAGE]**

(a) These two diagrams show the same graph of $y = f(x)$



(i) Sketch $y = f(x^2)$ on diagram (i) above, showing x intercepts and other key features of this graph.

3

(ii) Sketch $y = \log_e [f(x)]$ on diagram (ii) above, showing key features.

3

DETACH THIS PAGE AND ATTACH IT TO YOUR SOLUTIONS.

Question 3 continues on page 5

Question 3 (continued)

- (b) Find the **x-coordinates** of the points on the curve

$$2x^2 + 2xy + 3y^2 = 15$$

where the tangents to the curve are vertical.

3

- (c) (i) Sketch $y = x^2 - 2$ and $y = e^{-x}$ on the same number plane diagram. The diagram should be about one third of the page in size.

1

- (ii) Find the **x-coordinates** of the stationary points on $y = e^{-x}(x^2 - 2)$

2

- (iii) Hence, sketch the graph of $y = e^{-x}(x^2 - 2)$ on the same diagram as in (i), showing the x -intercepts and other key features of the graph.

3

End of Question 3

Question 4 (15 marks) **[START A NEW PAGE]**

- (a) An ellipse has the equation $\frac{x^2}{8} + \frac{y^2}{4} = 1$ and $P(x_1, y_1)$ is a point on this ellipse.
- (i) Find its eccentricity, the coordinates of its foci, S and S^1 , and the equations of its directrices. 3
- (ii) Prove that the sum of the distances SP and S^1P is independent of the position of P . 2
- (iii) Show that the equation of the tangent to the ellipse at P is $x_1 x + 2y_1 y = 8$. 3
- (iv) The tangent at $P(x_1, y_1)$ meets the directrix closest to S at T .
Prove that $\angle PST$ is a right angle. 3
- (b) The point $T\left(ct, \frac{c}{t}\right)$ lies on the hyperbola $xy = c^2$.
The normal at T meets the line $y = x$ at R .
Find the coordinates of R . 4

End of Question 4

Question 5 (15 marks) **[START A NEW PAGE]**

- (a) Given the polynomial

$$p(x) = ax^3 + bx^2 + cx + d$$

where a, b, c, d and β are integers and $p(\beta) = 0$:

- (i) Prove that
- β
- divides
- d
- 2

- (ii) Hence, or otherwise, prove that the polynomial equation

$$q(x) = 2x^3 - 5x^2 + 8x - 3 = 0 \quad \text{does not have an integer root.} \quad \text{2}$$

- (b) The numbers
- α, β
- and
- γ
- satisfy the equations

$$\alpha + \beta + \gamma = 0$$

$$\alpha^2 + \beta^2 + \gamma^2 = -2$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{-1}{10}$$

- (i) Find the values of
- $\alpha\beta + \beta\gamma + \alpha\gamma$
- and
- $\alpha\beta\gamma$
- 3

- (ii) Hence write down a cubic equation with roots
- α, β
- and
- γ

$$\text{in the form } ax^3 + bx^2 + cx + d = 0 \quad \text{1}$$

Question 5 continues on page 8

Question 5 (continued)

(c) The equation $x^3 + x^2 + 2x - 4 = 0$ has roots α , β , and γ .

(i) Evaluate $\alpha \beta \gamma$

1

(ii) Write an equation in the form

$$ax^3 + bx^2 + cx + d = 0$$

(A) with roots α^2 , β^2 and γ^2

3

(B) with roots $\alpha^2 \beta \gamma$, $\alpha \beta^2 \gamma$ and $\alpha \beta \gamma^2$

3

End of Question 5

Question 6 (15 marks) **[START A NEW PAGE]**

- (a) Find the volume of the solid generated when the area bounded by

$$y = 6 - x^2 - 3x \text{ and } y = 3 - x \text{ is revolved about the line } x = 3.$$

4

- (b) (i) By rewriting

$$\cos(n+2)x \text{ as } \cos\{(n+1)+1\}x,$$

and

$$\cos nx \text{ as } \cos\{(n+1)-1\}x,$$

$$\text{show that } \cos(n+2)x + \cos nx = 2\cos(n+1)x \cdot \cos(x)$$

1

- (ii) Hence prove that given
- $u_n = \int_0^\pi \frac{1 - \cos nx}{1 - \cos x} dx$

3where n is a positive integer or zero,

then,

$$\begin{aligned} u_{n+2} + u_n - 2u_{n+1} &= \int_0^\pi \frac{2\cos(n+1)x \cdot \{1 - \cos x\}}{1 - \cos x} dx \\ &= 0 \end{aligned}$$

- (iii) Evaluate
- u_0
- and
- u_1
- directly**
- , and hence evaluate
- u_2
- and
- u_3

3

- (iv) Also show that
- $\int_0^{\frac{\pi}{2}} \frac{\sin^2 3\theta}{\sin^2 \theta} d\theta = \frac{3\pi}{2}$

4**End of Question 6**

Question 7 (15 marks) **[START A NEW PAGE]**

- (a) The acceleration due to gravity at a point outside the Earth is inversely proportional to the square of the distance from the centre of the Earth, ie. $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{-k}{x^2}$

- (i) Neglecting air resistance, show that if a particle is projected vertically upwards with speed u from a point on the Earth's surface, its speed V in any position x is given by

$$V^2 = u^2 - 2gR^2 \left(\frac{1}{R} - \frac{1}{x} \right),$$

where R is the radius of the Earth, and g is the acceleration due to gravity at the Earth's surface.

3

- (ii) Show that the greatest height H , **above the Earth's surface**, reached by the particle is given by

$$H = \frac{u^2 R}{2gR - u^2}$$

2

- (iii) Prove that if the speed of projection exceeds 12 km/sec, the particle will escape the Earth's influence. (Take $R = 6400\text{km}$ and $g = 10\text{m/sec}^2$)

3

Question 7 continues on page 11

Question 7 (continued)

(b) Suppose that x is a positive number less than 1, and n is a non-negative integer.

(i) Explain why

$$1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots = \frac{1}{1+x}$$

and

$$\log_e (1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad 2$$

(ii) Hence, show that

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

and

$$\log\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots\right) \quad 2$$

(iii) By letting $x = \frac{1}{2m+1}$

$$(\alpha) \quad \text{Show that } \log\left(\frac{1+x}{1-x}\right) = \log\left(\frac{m+1}{m}\right) \quad 1$$

(β) Show that

$$\log\left(\frac{m+1}{m}\right) = 2\left(\frac{1}{2m+1} + \frac{1}{3(2m+1)^3} + \frac{1}{5(2m+1)^5} + \dots\right) \quad 1$$

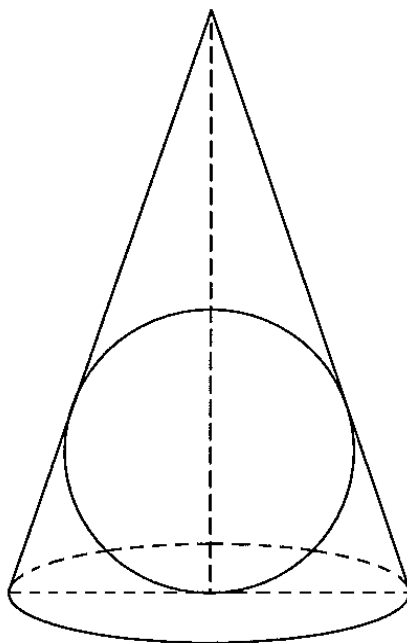
(iv) Use a result from (i), (ii) or (iii) to find a simple fraction which approximates the value of $\log_e(1.001)$ correctly to 9 decimal places. 1

End of Question 7

Question 8 (15 marks) **[START A NEW PAGE]**

- (a) You are trying to find the dimensions of the right circular cone of minimum volume which can be circumscribed about a **sphere of radius 20cm**, as shown below.

Let x cm = the radius of the base of the cone and let $(y + 20)$ cm = the altitude of the cone.



- (i) Prove that $x^2 = \frac{400(y + 20)}{y - 20}$ using similar triangles. 2
- (ii) Hence, find the dimensions of the cone which make its volume a minimum. 3

Question 8 continues on page 13

Question 8 (continued)

- (b) By using the formula for $\tan(\alpha - \beta)$ in terms of $\tan \alpha$ and $\tan \beta$, answer the following questions.

- (i) If $2x + y = \frac{\pi}{4}$, show that 2

$$\tan y = \frac{1 - 2 \tan x - \tan^2 x}{1 + 2 \tan x - \tan^2 x}$$

- (ii) Hence deduce that $\tan \frac{\pi}{8}$ is a root of the equation $t^2 + 2t - 1 = 0$ and find the exact value of $\tan\left(\frac{\pi}{8}\right)$ 3

- (c) For the series $S(x) = 1 + 2x + 3x^2 + \dots + (n + 1)x^n$, find $(1 - x)S(x)$ and hence find $S(x)$ 3

- (d) Find $\int_{-1}^1 x^2 \sin^7 x \, dx$, giving reasons. 2

End of Question 8

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Extension 2 Trial HSC 2011

① (a) (i) $-\int \frac{dx}{x^2+2x-3} = -\int \frac{dx}{(x+3)(x-1)}$ (ii) $\sin^{-1}\left(\frac{x+1}{2}\right) + \int \frac{2x}{\sqrt{3-2x-x^2}}$

Let $\frac{1}{(x+3)(x-1)} = \frac{A}{x+3} + \frac{B}{x-1}$ $= \sin^{-1}\left(\frac{x+1}{2}\right) + \int 2x(3-2x-x^2)^{-\frac{1}{2}} dx$

$\therefore 1 = A(x-1) + B(x+3)$ $= \sin^{-1}\left(\frac{x+1}{2}\right) + \int \frac{2x+2-2}{\sqrt{3-2x-x^2}} dx$

Let $x=1 \therefore 1 = 4B \therefore B = \frac{1}{4}$ $= \sin^{-1}\left(\frac{x+1}{2}\right) - 2 \int \frac{1}{\sqrt{3-2x-x^2}} dx - \int \frac{2x-2}{\sqrt{3-2x-x^2}} dx$

Let $x=-3 \therefore 1 = -4A \therefore A = -\frac{1}{4}$ $= \sin^{-1}\left(\frac{x+1}{2}\right) - 2 \sin^{-1}\left(\frac{x+1}{2}\right) - \int (2x-2)(3-2x-x^2)^{-\frac{1}{2}} dx$

$\therefore$ Answer: $-\int \frac{1}{x-1} - \frac{1}{x+3} dx = -\sin^{-1}\left(\frac{x+1}{2}\right) - 2 \sin^{-1}\left(\frac{x+1}{2}\right) - \int (2x-2)(3-2x-x^2)^{-\frac{1}{2}} dx$

$= -\frac{1}{4} \int \frac{1}{x-1} - \frac{1}{x+3} dx = -\sin^{-1}\left(\frac{x+1}{2}\right) - 2 \sqrt{3-2x-x^2} + C$

$= -\frac{1}{4} \ln \left| \frac{x-1}{x+3} \right| + C$ (c) $\int \frac{1}{\sqrt{x^2+a^2}} dx$

(ii) $-\int \frac{2+x}{x^2+2x-3} dx = x\sqrt{x^2+a^2} - \int x + \frac{1}{2} \frac{2x}{\sqrt{x^2+a^2}} dx$

$= -\frac{1}{2} \ln \left| \frac{x-1}{x+3} \right| - \int \frac{x}{x^2+2x-3} dx = x\sqrt{x^2+a^2} - \int \frac{x^2}{\sqrt{x^2+a^2}} dx$

$= -\frac{1}{2} \ln \left| \frac{x-1}{x+3} \right| - \int \frac{x+1-1}{x^2+2x-3} dx = x\sqrt{x^2+a^2} - \int \frac{x^2+a^2-a^2}{\sqrt{x^2+a^2}} dx$

$= -\frac{1}{2} \ln \left| \frac{x-1}{x+3} \right| - \frac{1}{2} \int \frac{2x+2}{x^2+2x-3} dx + \int \frac{1}{x^2+2x-3} dx = x\sqrt{x^2+a^2} - \int \sqrt{x^2+a^2} dx + \int \frac{a^2}{\sqrt{x^2+a^2}} dx$

$= -\frac{1}{2} \ln \left| \frac{x-1}{x+3} \right| - \frac{1}{2} \ln |x^2+2x-3| + \frac{1}{4} \ln \left| \frac{x-1}{x+3} \right| + C \therefore 2 \int \sqrt{x^2+a^2} dx = x\sqrt{x^2+a^2} + a^2 \ln(x+\sqrt{x^2+a^2})$

$= -\frac{1}{4} \ln \left| \frac{x-1}{x+3} \right| - \frac{1}{2} \ln |x^2+2x-3| + C \therefore \sin^{-1}\left(\frac{x+1}{2}\right) + C$

(b) (i) $\int \frac{dx}{\sqrt{x^2+2x-3}} = \sin^{-1}\left(\frac{x+1}{2}\right) + C$

$= \int \frac{dx}{\sqrt{(x+1)^2-4}} \therefore \int \sqrt{x^2+a^2} dx = \frac{x\sqrt{x^2+a^2} + a^2 \ln(x+\sqrt{x^2+a^2})}{2} + C$

$= \int \frac{dx}{\sqrt{4-(x+1)^2}}$

② (a) (i) $z^2 = 2 \cos\left(\frac{\pi}{4} + 2k\pi\right)$ (c) $(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$

$\therefore z = \sqrt[5]{2} \cos\left(\frac{\pi}{12} + \frac{2k\pi}{5}\right)$ $\therefore \cos^5\theta + 5\cos^4\theta \sin\theta + 10\cos^3\theta \sin^2\theta - 10\cos^2\theta \sin^3\theta + 5\cos\theta \sin^4\theta + i\sin^5\theta = \cos 5\theta + i\sin 5\theta$

$z = \sqrt[5]{2} \cos\left(\frac{\pi(1+8k)}{12}\right)$ Equating real, $\cos 5\theta = \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta) + 5\cos\theta(1-\cos^2\theta)(1-\cos^2\theta)$

(ii) For $z^2 - (5+2i) = 0$, $\cos 5\theta = \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

Sum of roots = 0 $\therefore \sqrt[5]{2} \left(\cos \frac{\pi}{12} + \cos \frac{7\pi}{12} + \cos \frac{11\pi}{12} \right) = 0$ $\therefore \cos \frac{\pi}{12} + \cos \frac{7\pi}{12} + \cos \frac{11\pi}{12} = 0$

$\therefore \cos \frac{\pi}{12} + \cos \frac{7\pi}{12} + \cos \frac{11\pi}{12} = 0$ (ii) $16x^5 - 20x^3 + 5x = \frac{1}{2}$

$\therefore$ real sum & imaginary sum = 0 $\therefore \cos \frac{\pi}{12} + \cos \frac{7\pi}{12} + \cos \frac{11\pi}{12} = 0$ Let $x = \cos\theta$

(b) (i) If $\frac{z-i}{z-2}$ is purely real, $\therefore$ solve $16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = \frac{1}{2}$

$\arg\left(\frac{z-i}{z-2}\right) = 0 \text{ or } \pi$ $\therefore \cos 5\theta = \frac{1}{2}$

$\therefore \arg(z-i) - \arg(z-2) = 0 \text{ or } \pi$ $5\theta = \pm \frac{\pi}{3} + 2k\pi$

$\theta = \pm \frac{\pi}{15} + \frac{2k\pi}{5}$ $\therefore x = \cos\left(\frac{2k\pi}{5} \pm \frac{\pi}{15}\right)$

Since 5 sides: $x = \cos \frac{\pi}{15}, \cos \frac{7\pi}{15}, \cos \frac{11\pi}{15}, \cos \frac{13\pi}{15}, \cos \frac{17\pi}{15}$

$= \cos \frac{\pi}{15}, \frac{1}{2}, \cos \frac{7\pi}{15}, \cos \frac{11\pi}{15}, \cos \frac{13\pi}{15}$

(ii) $\arg\left(\frac{z-i}{z-2}\right) = \pm \frac{\pi}{2}$ (iii) Product of roots of $32x^5 - 40x^3 + 10x - 1 = 0$

$\therefore \arg(z-i) - \arg(z-2) = \pm \frac{\pi}{2}$ $\frac{1}{2} \cos \frac{\pi}{15} \cos \frac{7\pi}{15} \cos \frac{11\pi}{15} \cos \frac{13\pi}{15}$

Note: $32x^5 - 40x^3 + 10x - 1 = 0$ has product of roots = $-\left(\frac{-1}{32}\right) = \frac{1}{32}$

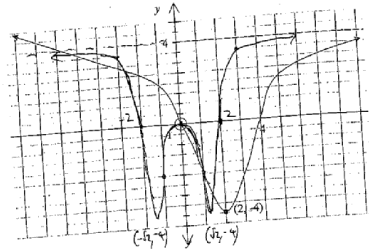
$\therefore \frac{1}{2} \cos \frac{\pi}{15} \cos \frac{7\pi}{15} \cos \frac{11\pi}{15} \cos \frac{13\pi}{15} = \frac{1}{32}$

$\therefore \cos \frac{\pi}{15} \cos \frac{7\pi}{15} \cos \frac{11\pi}{15} \cos \frac{13\pi}{15} = \frac{1}{16}$

(3) (a)

(i) $y = f(x^2)$

(ii) see below



(b) $4x + 2\left(y + x \frac{dy}{dx}\right) + 6y \frac{dy}{dx} = 0$

$4x + 2y + 2x \frac{dy}{dx} + 6y \frac{dy}{dx} = 0$

$\frac{dy}{dx}(2x + 6y) = -(4x + 2y)$

$\therefore \frac{dy}{dx} = -\frac{4x + 2y}{2x + 6y} = -\frac{2x + y}{x + 3y}$

For vertical tangents, $x + 3y = 0$

$\therefore x = -3y$

$\therefore 2(-3y)^2 + 2(-3y)y + y^2 = 15$

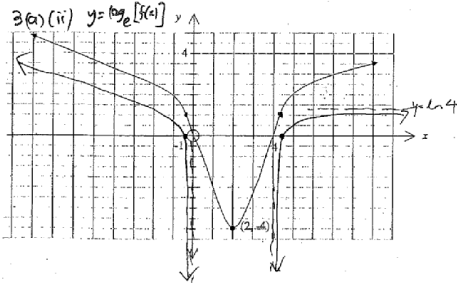
$(8y^2 - 6y^2 + 3y^2) = 15$

$15y^2 = 15 \therefore y = \pm 1$

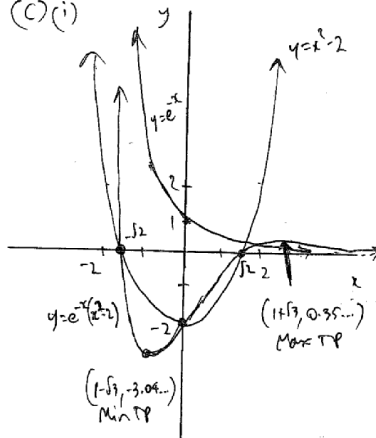
$y = 1 \rightarrow x = -3$

$y = -1 \rightarrow x = 3$

$\therefore x$ coordinates are ± 3



(c) (i)



(ii) $\frac{dy}{dx} = -e^{-x}(x^2 - 2) + 2xe^{-x}$

$= e^{-x}[2x - x^2 + 2]$

$= e^{-x}[x^2 - 2x - 2]$

$= -e^{-x}[(x-1)^2 - 3] = 0$ for stat pt

$\therefore (x-1)^2 = 3$

$x - 1 = \pm\sqrt{3}$

$x = 1 \pm \sqrt{3}$

$\left[1 \pm 2.732 \dots \right]$

(14) (a) $\frac{x^2}{8} + \frac{y^2}{4} = 1$

(i) $a = 2\sqrt{2}$
 $b = 2$

$b^2 = a^2(1 - e^2)$

$4 = 8(1 - e^2)$

$1 - e^2 = \frac{1}{2}$

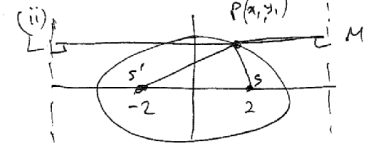
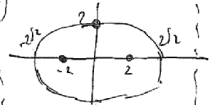
$e^2 = \frac{1}{2}$

$e = \frac{1}{\sqrt{2}}$

$ae = 2$

$\frac{a}{e} = 4$

Foci $(\pm 2, 0)$
Directrices $x = \pm 4$



$x = -4$

Construct LPM straight horizontal line

Now $\frac{PS}{PM} = \frac{1}{\sqrt{2}}$ & $\frac{PS'}{PL} = \frac{1}{\sqrt{2}}$

$\therefore PM = \sqrt{2}PS$ & $PL = \sqrt{2}PS'$

and $PM + PL = 8$

$\therefore 8 = \sqrt{2}(PS + PS')$

$\therefore PS + PS' = \frac{8}{\sqrt{2}} = 4\sqrt{2}$ which is independent of P 's position

(iii) $\frac{x}{4} + \frac{y}{2} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{x}{2y} = -\frac{x}{2y}$ $\therefore m = -\frac{x}{2y}$

$\therefore y - y_1 = -\frac{x_1}{2y_1}(x - x_1)$

$2yy_1 - 2y_1^2 = x_1^2 - xx_1$

$xx_1 + 2yy_1 = x_1^2 + 2y_1^2$

$\therefore \frac{xx_1}{8} + \frac{yy_1}{4} = \frac{x_1^2}{8} + \frac{y_1^2}{4}$

Now $\frac{x_1^2}{8} + \frac{y_1^2}{4} = 1$

From eqn of ellipse

$\therefore \frac{xx_1}{8} + \frac{yy_1}{4} = 1$

$\therefore xx_1 + 2yy_1 = 8$

(iv) For T:

$4x_1 + 2yy_1 = 8$

$2yy_1 = 8 - 4x_1$

$y = \frac{8 - 4x_1}{2y_1} = \frac{4 - 2x_1}{y_1}$

$\therefore T\left(4, \frac{4 - 2x_1}{y_1}\right)$

$M_{PS} = \frac{y_1 - 0}{x_1 - 2} = \frac{y_1}{x_1 - 2}$

$M_{TS} = \frac{\frac{4 - 2x_1}{y_1} - 0}{\frac{4 - 2x_1}{y_1} - 2} = \frac{4 - 2x_1}{2y_1}$

$M_{PS} \times M_{TS} = \frac{y_1}{x_1 - 2} \times \frac{4 - 2x_1}{2y_1}$

$= \frac{y_1}{2y_1} \times \frac{2(2 - x_1)}{x_1 - 2}$

$= \frac{-(x_1 - 2)}{x_1 - 2} = -1$

$\therefore \angle PST$ is a right angle

(b) $T\left(\frac{c}{k}, \frac{c}{k}\right)$

$y = \frac{c}{x} \therefore \frac{dy}{dx} = -\frac{c}{x^2}$

$\therefore m_{normal} = \frac{(cd)^2}{c^2} = d^2$

$\therefore y - \frac{c}{k} = d^2(x - \frac{c}{k})$ is normal eqn

For R: solve simultaneously with $y = \frac{c}{x}$

$\therefore x - \frac{c}{k} = d^2(x - \frac{c}{k})$

$kx - c = k^3(x - \frac{c}{k})$

$x(k - k^3) = c - c k^2 = c(1 - k^2)$

$x = \frac{c(1 - k^2)}{k(1 - k^2)} = \frac{c}{k}$

$\therefore R\left(\frac{c}{k}(1 + k^2), \frac{c}{k}(1 + k^2)\right)$

$R\left(\frac{c}{k} + ck^2, \frac{c}{k} + ck^2\right)$

$= \text{sum of ordinates of } T \text{ and } S$

⑤ (a) (i) $p(\beta) = 0$

$\therefore a\beta^3 + b\beta^2 + c\beta + d = 0$

$\therefore a\beta^3 + b\beta^2 + c\beta = -d$

$\therefore \beta \cdot (a\beta^2 + b\beta + c) = -d$

Note: if $\beta = 0$, that means $\beta = 0$, which still satisfies
Since $a, b, c \neq \beta$ are integers, β dividing d

LHS is an integer

$\therefore \frac{-d}{\beta}$ is an integer

Since $d \neq \beta$ are integers,
 β must divide exactly into d

(ii) Need to test integers $\pm 1, \pm 3$

$q(1) = 2 \quad q(3) = 30$

$q(-1) = -18 \quad q(-3) = -126$

$\therefore \pm 1, \pm 3$ not roots
since they are only factors of 3,
No integer root

(b) (i) $\alpha\beta + \beta\delta + \gamma\delta = \frac{(\alpha + \beta + \delta)^2 - (\alpha^2 + \beta^2 + \delta^2)}{2}$

$= \frac{0^2 - (-2)}{2} = 1$

$\alpha\beta\delta = \frac{\alpha + \beta + \delta - \gamma}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\delta}} = \frac{1}{-\frac{1}{10}} = -10$

(ii) Let $a = 1$

$\therefore c = 1 \quad (\alpha + \beta + \delta + \gamma)$

$d = 10 \quad (-\alpha\beta\delta)$

$b = 0 \quad (-\alpha\beta\delta + \gamma)$

$\therefore x^3 + 1 + 10 = 0$

(c) (i) $\alpha\beta\gamma = 4$

(ii) (a) Let $y = x^2 \therefore x = \sqrt{y}$

$\therefore y\sqrt{y} + y + 2\sqrt{y} - 4 = 0$

$\sqrt{y}(y + 1) = 4 - y$

$y(y^2 + 4y + 4) = 16 - 8y + y^2$

$y^3 + 3y^2 + 12y - 16 = 0$

$\therefore x^3 + 3x^2 + 12x - 16 = 0$

(b) Roots: $\alpha(-\beta\delta), \beta(-\alpha\delta), \delta(-\alpha\beta)$

$= 4\alpha, 4\beta, 4\delta$

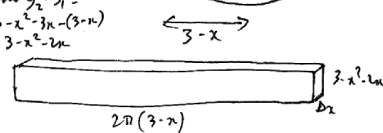
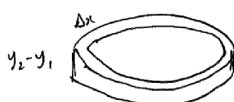
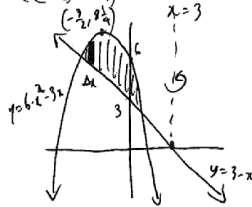
$\therefore$ Let $y = 4x \therefore x = \frac{y}{4}$

$\therefore \frac{y^3}{64} + \frac{y^2}{16} + \frac{y}{4} - 4 = 0$

$y^3 + 4y^2 + 32y - 256 = 0$

$\therefore x^3 + 4x^2 + 32x - 256 = 0$

(6) $y = -(x^2 + 3x - 6)$
 $y = -\left(\left(x + \frac{3}{2}\right)^2 - \frac{9}{4} - 6\right) = -\left(x + \frac{3}{2}\right)^2 + \frac{21}{4}$



Points of intersection: $3 - x^2 - 2x = 0$
 $\therefore x^2 + 2x - 3 = 0$
 $(x + 3)(x - 1) = 0$
 $x = -3$ or 1

$\therefore V = 2\pi \int_{-3}^1 (3-x)(3-x^2-2x) dx$
 $= 2\pi \int_{-3}^1 (9 - 3x^2 - 6x - 3x^3 + 2x^2) dx$
 $= 2\pi \int_{-3}^1 (9 - x^2 - 4x - 3x^3) dx = 2\pi \left[9x - \frac{x^3}{3} - 2x^2 - \frac{3x^4}{4} \right]_{-3}^1 = \frac{25\pi}{3}$

(b) (i) $\cos(n+1)x + \cos nx =$
 $\cos(n+1)x \cos x - \sin(n+1)x \sin x$
 $+ \cos(n+1)x \cos x + \sin(n+1)x \sin x$
 $= 2\cos(n+1)x \cos x$

(ii) $U_{n+2} + U_n - 2U_{n+1} =$

$\int_0^\pi \frac{1 - \cos(n+2)x + 1 - \cos nx - 2(1 - \cos(n+1)x)}{1 - \cos x} dx$

$= \int_0^\pi \frac{2 - \cos(n+2)x - \cos nx - 2 + 2\cos(n+1)x}{1 - \cos x} dx$

$= \int_0^\pi \frac{2\cos(n+1)x - \cos(n+2)x - \cos nx}{1 - \cos x} dx$

$= \int_0^\pi \frac{2\cos(n+1)x - (2\cos(n+1)x \cos x)}{1 - \cos x} dx$

$= \int_0^\pi \frac{2\cos(n+1)x [1 - \cos x]}{1 - \cos x} dx$

$= \int_0^\pi 2\cos(n+1)x dx$

$= \left[\frac{2\sin(n+1)x}{n+1} \right]_0^\pi$

$= \frac{2}{n+1} [\sin(n+1)\pi - \sin(n+1)0]$

$= \frac{2}{n+1} [0 - 0] \text{ since } \sin k\pi = 0 \text{ for } k \text{ integer}$

(ii) $U_0 = \int_0^\pi 0 dx = 0$

$U_1 = \int_0^\pi 1 dx = [x]_0^\pi = \pi$

$U_2 + U_0 - 2U_1 = 0$

$\therefore U_2 = 2\pi$

$U_3 + U_1 - 2U_2 = 0$

$\therefore U_3 = 3\pi$

(iv)

To change limits,
let's try this substitution:
let $x = 2\theta$
 $\therefore$ when $\theta = 0, x = 0$
when $\theta = \frac{\pi}{2}, x = \pi$

$\therefore \int_0^\pi \frac{\sin^2 3\theta}{\sin \theta} d\theta$

Now if $x = 2\theta$,
 $\cos x = \cos 2\theta = 2\cos^2 \theta - 1$
 $= 1 - 2\sin^2 \theta$

$\therefore \sin^2 \theta = \frac{1 - \cos x}{2}$

$\therefore \cos 3x = \cos 6\theta = 1 - 2\sin^2 3\theta$

$\therefore \sin^2 3\theta = \frac{1 - \cos 3x}{2}$

$\therefore \frac{1}{2} \int_0^\pi \frac{1 - \cos 3x}{1 - \cos x} dx = \frac{1}{2} \int_0^\pi \frac{1 - \cos 3x}{1 - \cos x} dx$
 $= \frac{1}{2} U_3 = \frac{3\pi}{2}$

8(c) $(1-x) f(x) = (1 + 2x + 3x^2 + 4x^3 + \dots + (n+1)x^n)$
 $x + 2x^2 + 3x^3 + 4x^4 + \dots + nx^n + (n+1)x^{n+1}$
 $= (1 + x + x^2 + x^3 + \dots + x^n) \cdot (n+1)x^{n+1}$
 $\therefore S(x) = \frac{1 - x^{n+1}}{1 - x} - (n+1)x^{n+1}$
 $= \frac{1 - x^{n+1} - (n+1)x^{n+1}(1 - x)}{(1 - x)^2}$
 $= \frac{1 - x^{n+1} - (n+1)x^{n+1} + (n+1)x^{n+2}}{(1 - x)^2}$
 $= \frac{(n+1)x^{n+2} - (n+1)x^{n+1} + 1}{(1 - x)^2} = \frac{1 + (n+1)x^{n+2} - (n+1)x^{n+1}}{(1 - x)^2}$

⑦ (i) $\frac{1}{2} u^2 = \int -kx^2 dx$

$\frac{1}{2} u^2 = \frac{k}{x} + C$

When $x = R, v = u$

$\therefore \frac{1}{2} u^2 = \frac{k}{R} + C$

$\therefore C = \frac{u^2}{2} - \frac{k}{R}$

$\therefore \frac{1}{2} v^2 = \frac{k}{x} + \frac{u^2}{2} - \frac{k}{R}$

$\therefore v^2 = \frac{2k}{x} + u^2 - \frac{2k}{R}$

$v^2 = u^2 + 2k \left(\frac{1}{x} - \frac{1}{R} \right)$

Now to find k :

When $x = R, \ddot{x} = -g$

$\therefore -g = \frac{-k}{R^2} \therefore k = gR^2$

$\therefore v^2 = u^2 + 2(gR^2) \left(\frac{1}{x} - \frac{1}{R} \right)$

$v^2 = u^2 - 2gR^2 \left(\frac{1}{R} - \frac{1}{x} \right)$

(ii) Greatest height when $v = 0$

$\therefore u^2 = 2gR^2 \left(\frac{1}{R} - \frac{1}{x} \right)$

$\frac{u^2}{2gR^2} = \frac{1}{R} - \frac{1}{x}$

$\frac{1}{x} = \frac{1}{R} - \frac{u^2}{2gR^2}$

$\frac{1}{x} = \frac{2gR - u^2}{2gR^2}$

$\therefore x = \frac{2gR^2}{2gR - u^2}$

But x is distance from centre of earth.

$\therefore H = \frac{2gR^2}{2gR - u^2} - R = \frac{2gR^2 - 2gR^2 + Ru^2}{2gR - u^2}$

$= \frac{u^2 R}{2gR - u^2}$

(iii) To escape, $H \rightarrow \infty$

i.e. $2gR \rightarrow u^2$

$\therefore 902 \times 6400 \rightarrow u^2$ [units in km]

$128 \rightarrow u^2$

$\therefore u \rightarrow \sqrt{128} \approx 11.3 \dots \text{km/sec}$

$\therefore$ If particle exceeds 12 km/sec, will escape!

(b)(i) $1 - x + x^2 - x^3 + \dots + (-1)^n x^n$ is an infinite series, $a = 1, r = -x$. Note $|r| < 1$ since $0 < x < 1$

$\therefore$ The sum $= \frac{1}{1 - (-x)} = \frac{1}{1+x}$

Integrating both sides:

$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \ln(1+x) + C$

When $x = 0, 0 = 0 + C \therefore C = 0$

$\therefore x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \ln(1+x)$

(ii) Replace x by $-x$

$\therefore \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$

$\therefore \ln\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$

$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

$-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$

$= 2x + 2\frac{x^2}{2} + 2\frac{x^3}{3} + \dots$

$= 2\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \dots\right)$

(iii) (a) $\ln\left(1 + \frac{1}{2m+1}\right) = \ln\left(\frac{2m+2}{2m+1}\right) = \ln\left(\frac{2m+2}{2m+1}\right)$

$= \ln\left(\frac{m+1}{m}\right)$

(b) $2\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \dots\right) = 2\left(\frac{1}{2m+1} + \frac{1}{3(2m+1)^2} + \dots\right)$

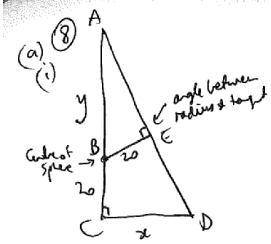
$\therefore$ Prove

(iv) Let $m = 1000 \therefore \ln\left(1 + \frac{1}{2001}\right) =$

$2\left(\frac{1}{2001} + \frac{1}{3(2001)^2} + \frac{1}{5(2001)^3} + \frac{1}{7(2001)^4} + \dots\right)$

$= 9.995003331 \times 10^{-4}$ To 4 d.p., $\frac{2}{2001}$

$= 0.000999500$ does the trick!



$\triangle ABE \parallel \triangle ADC$ (equiangular)
The $\angle A$ is common & $\angle AEB = \angle ACD = 90^\circ$

$\therefore \frac{20}{x} = \frac{y}{\sqrt{x^2 + (y+20)^2}}$

$400(x^2 + (y+20)^2) = x^2 y^2$

$400x^2 + 400y^2 + 16000y + 160000 = x^2 y^2$

$x^2(y^2 - 400) = 400y^2 + 16000y + 160000$

$x^2(y^2 - 400) = 400(y^2 + 40y + 400)$

$x^2(y-20)(y+20) = 400(y+20)^2$

$x^2 = \frac{400(y+20)}{y-20}$ [which $\neq 0$]

(ii)

$V = \frac{1}{3} \pi x^2 (y+20)$

$V = \frac{1}{3} \pi \frac{400(y+20)(y+20)}{y-20}$

$V = \frac{400\pi}{3} \frac{(y+20)^2}{y-20}$

$\frac{dV}{dy} = \frac{400\pi}{3} \left[\frac{2(y+20)(y-20) - (y+20)^2}{(y-20)^2} \right]$

$= \frac{400\pi}{3} \left[\frac{(y+20)(2y-40-y-20)}{(y-20)^2} \right]$

$= \frac{400\pi}{3} \left[\frac{(y+20)(y-60)}{(y-20)^2} \right]$

For stat pt, $(y+20)(y-60) = 0$
 $\therefore y = -20$ or 60 .

cone has no height

Check for minimum:

y	50	60	70
$\frac{dV}{dy}$	$\frac{400\pi(70-60)}{3 \cdot 20^2}$	0	$\frac{400\pi(90-60)}{3 \cdot 30^2}$

$\therefore$ Min Volume

$\therefore$ dimensions: Height = 80 cm

$x^2 = \frac{400 \times 80}{40} = 800$
 $\therefore$ Radius = $\sqrt{800}$ cm = $20\sqrt{2}$ cm

(b)(i) $y = \frac{\pi}{4} - 2x$
 $\therefore \tan y = \frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{1 - \frac{2 \tan x}{1 + \tan^2 x}}{1 + \frac{2 \tan x}{1 + \tan^2 x}}$

$= \frac{1 - \tan^2 x - 2 \tan x}{1 - \tan^2 x + 2 \tan x}$

(ii)

Let $x = \frac{\pi}{8} \therefore y = 0$

$\therefore \tan 0 = 0 = \frac{1 - 2 \tan \frac{\pi}{8} - \tan^2 \frac{\pi}{8}}{1 + 2 \tan \frac{\pi}{8} - \tan^2 \frac{\pi}{8}}$

$\therefore 1 - 2 \tan \frac{\pi}{8} - \tan^2 \frac{\pi}{8} = 0$

$\therefore \tan^2 \frac{\pi}{8} + 2 \tan \frac{\pi}{8} - 1 = 0$

$\therefore$ For $x^2 + 2x - 1 = 0, x = \tan \frac{\pi}{8}$ is a soln

$\therefore \tan \frac{\pi}{8} = \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} = -1 \pm \sqrt{2}$

(c) Previous since $\tan \frac{\pi}{8}$ is +ve, $\tan \frac{\pi}{8} = \sqrt{2} - 1$

(d) Let $f(x) = x^2 \sin^2 x$

$\therefore f(-x) = (-x)^2 \sin^2(-x)$

$= x^2 (-\sin x)^2$

$= x^2 (-1) \sin^2 x$

$= -x^2 \sin^2 x$

$= -f(x)$

$\therefore f(-x) = -f(x) \therefore$ Odd function

since $x \sin^2 x$ is odd about y-axis